



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Scott Allan
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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Objective:** Predict whether the Falcon 9 first stage will successfully land, using publicly available launch data, to support reusability-driven cost estimation.
- **Methodologies:**
 - Collected launch data via SpaceX REST API and complemented records using Wikipedia web scraping.
 - Processed and wrangled data (cleaning, feature formatting, missing value handling) and created a binary landing success label (0/1).
 - Conducted EDA and data visualisation to identify trends across payload mass, orbit, launch site and flight number.
 - Built and tuned classification models using GridSearchCV cross-validation (Logistic Regression, SVM, Decision Tree, KNN).
 - Evaluated models using test accuracy and confusion matrices, selecting the most reliable performer.
- **Results:**
 - EDA showed that landing success improves over time and is influenced by orbit type, payload mass and launch site.
 - All models achieved reasonable predictive performance; the strongest models achieved ~0.83 test accuracy, while Decision Tree performed lower (~0.72).
 - The best-performing classification approach provided a robust method to estimate landing success probability, supporting improved reuse likelihood and launch cost/pricing decisions.

Introduction

- **Context:** SpaceX has dramatically reduced launch costs by reusing the Falcon 9 first stage, giving it a strong competitive advantage.
- **Challenge:** For a new launch company (Space Y), launch pricing depends heavily on whether the first stage can be recovered and reused.
- **Problems to be answered:**
 1. What factors (e.g. payload mass, orbit type, launch site) influence whether the Falcon 9 first stage successfully lands?
 2. Can we predict first-stage landing success using publicly available launch data?
 3. Which machine learning model provides the most accurate and reliable predictions?
 4. How can these predictions be used to estimate launch cost and support pricing decisions?
- **Goal:** Develop a data-driven model and dashboards to support cost estimation and decision-making for Space Y.

Section 1

Methodology

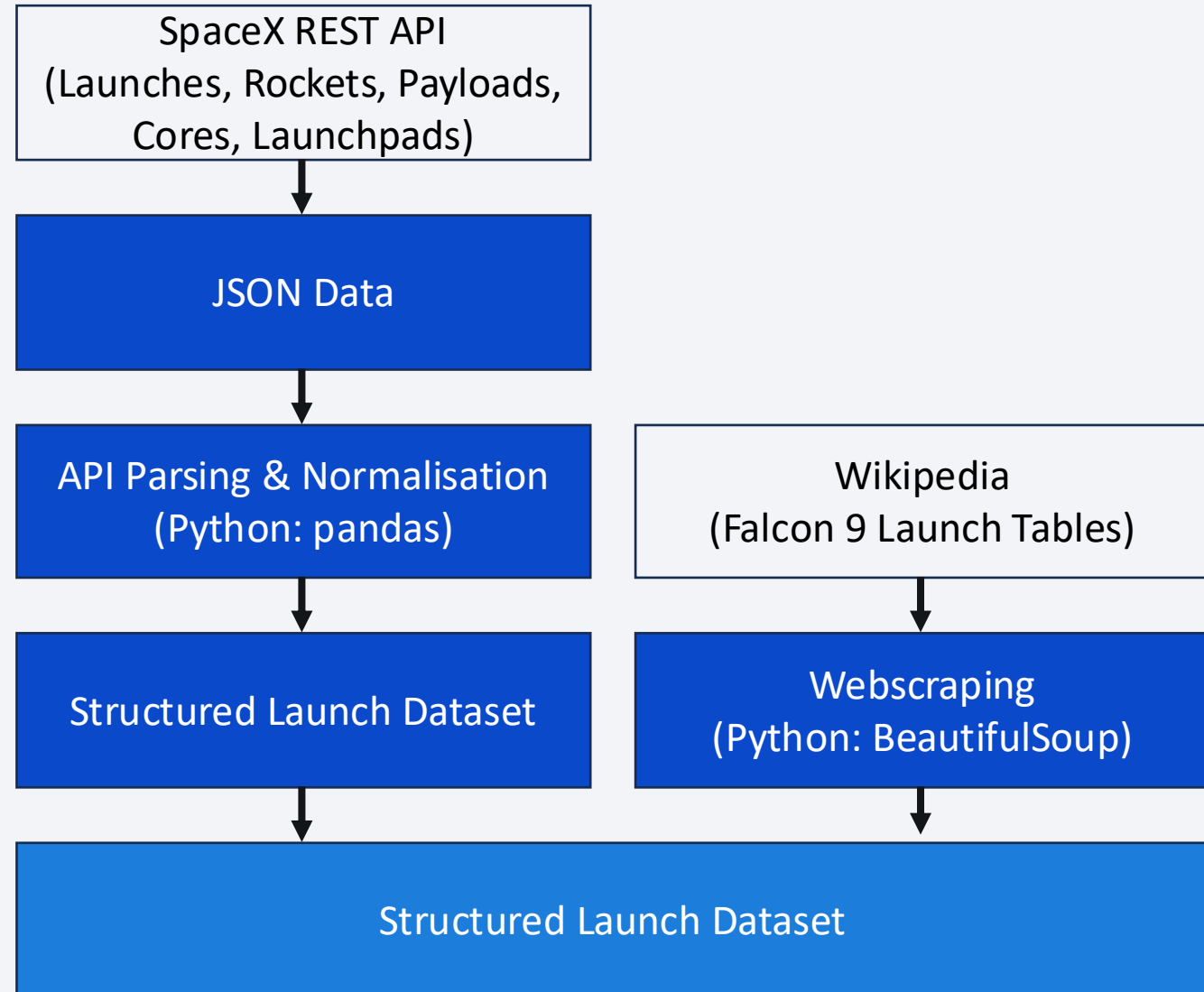
Methodology

Executive Summary

- Data collection methodology:
 - Querying the SpaceX REST API to retrieve structured launch, rocket, payload, core & launch site data
 - Web scraping historical Falcon 9 launch records (Wikipedia) using BeautifulSoup to confirm landing outcomes
- Perform data wrangling
 - Data was cleaned and structured by handling missing values, standardising variable types, aggregating launch attributes and converting landing outcomes into a binary classification label for model training
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Classification models (logistic regression, SVM, decision trees and kNN were trained and tuned using cross-validation, with accuracy and confusion matrices used to identify the best model.

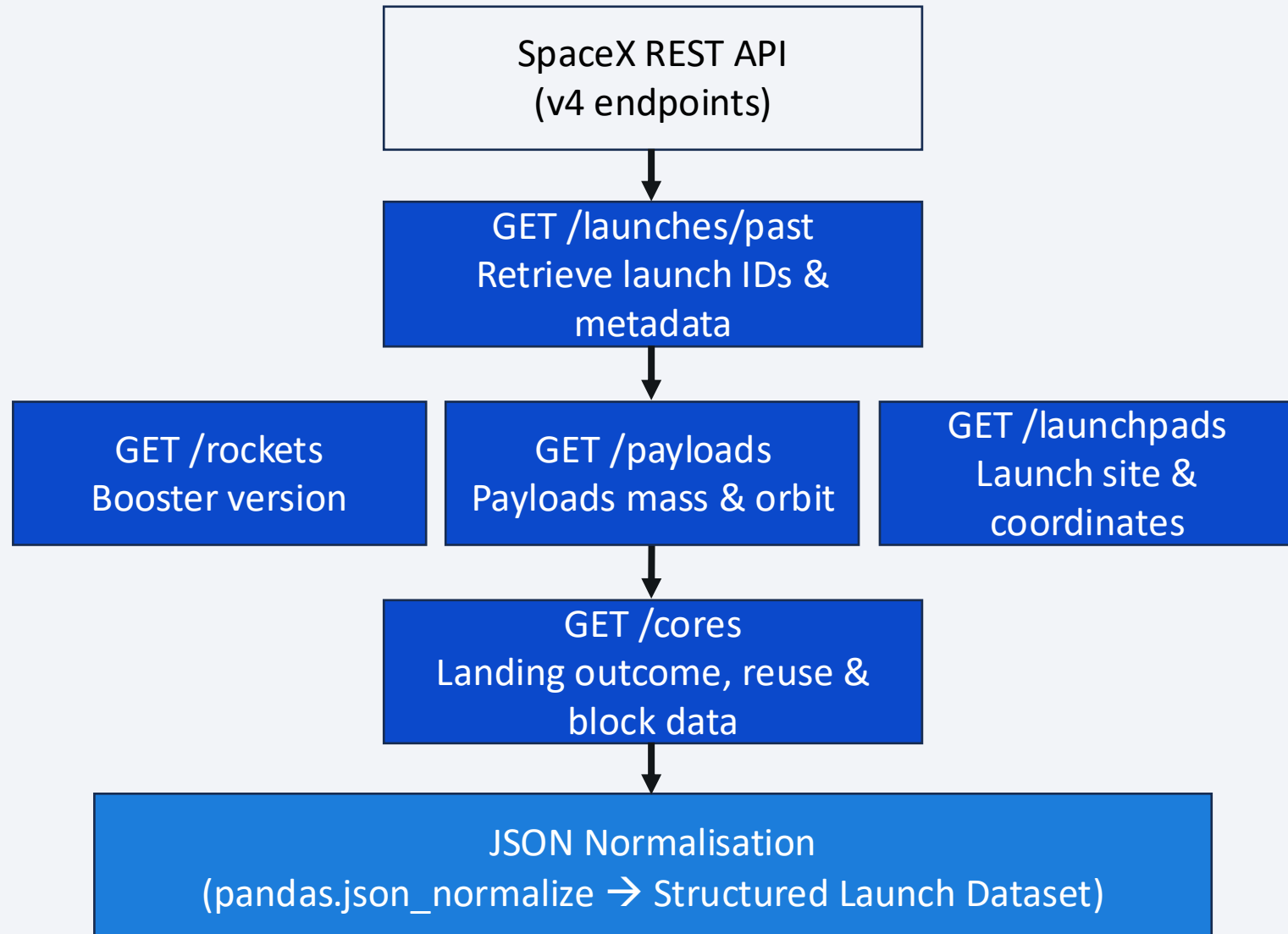
Data Collection

- Data sets were collected from the following sources:
 - **Primary source:** Querying the SpaceX REST API to retrieve structured launch, rocket, payload, core & launch site data
 - **Secondary source:** Web scraping historical Falcon 9 launch records (Wikipedia) using BeautifulSoup to confirm landing outcomes



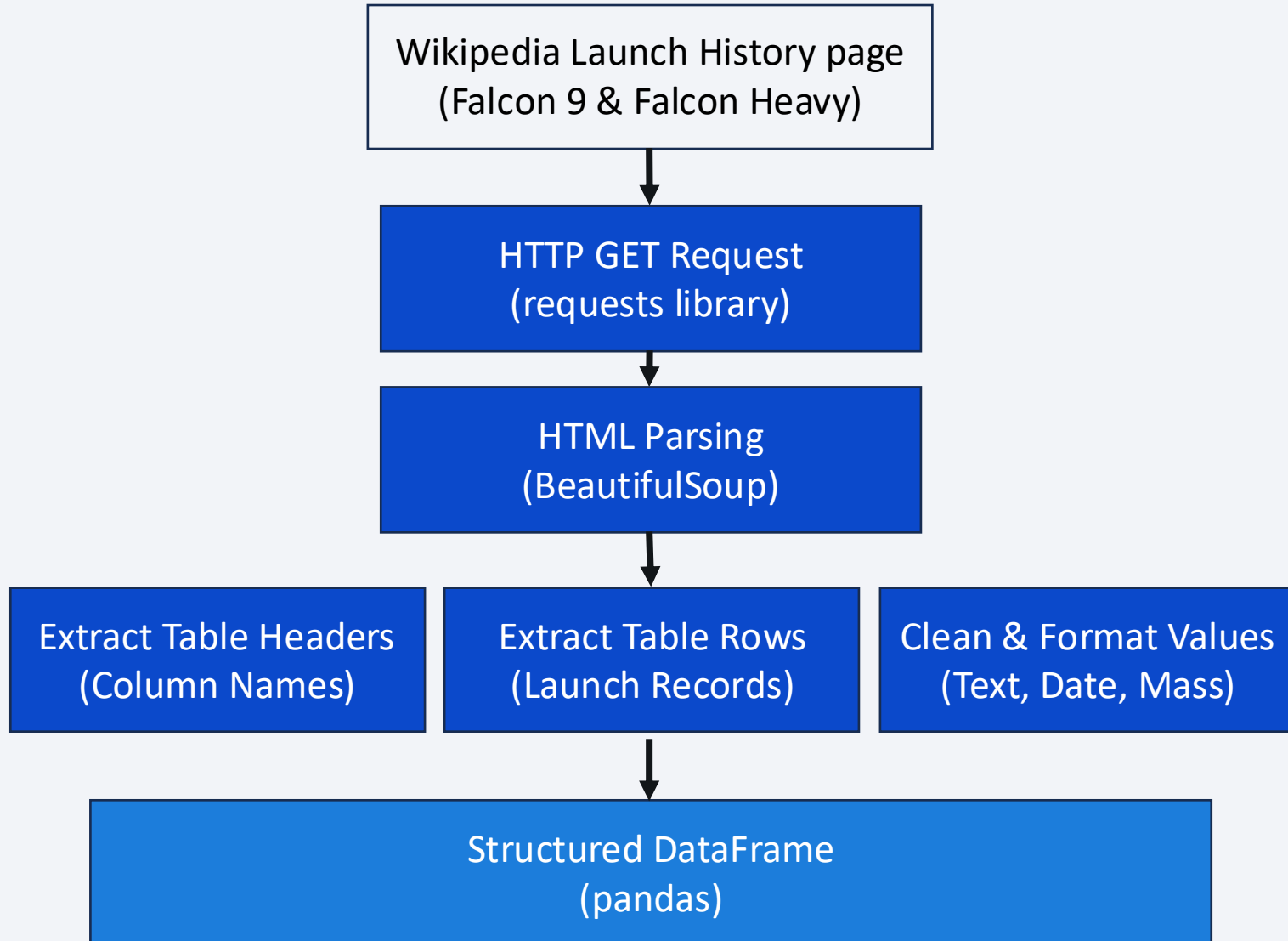
Data Collection – SpaceX API

- Github URL of SpaceX API notebook:
- <https://github.com/sja08/Capstone1/blob/main/10.1%20lab%20-%20data%20collection%20API%20and%20data%20wrangling.ipynb>



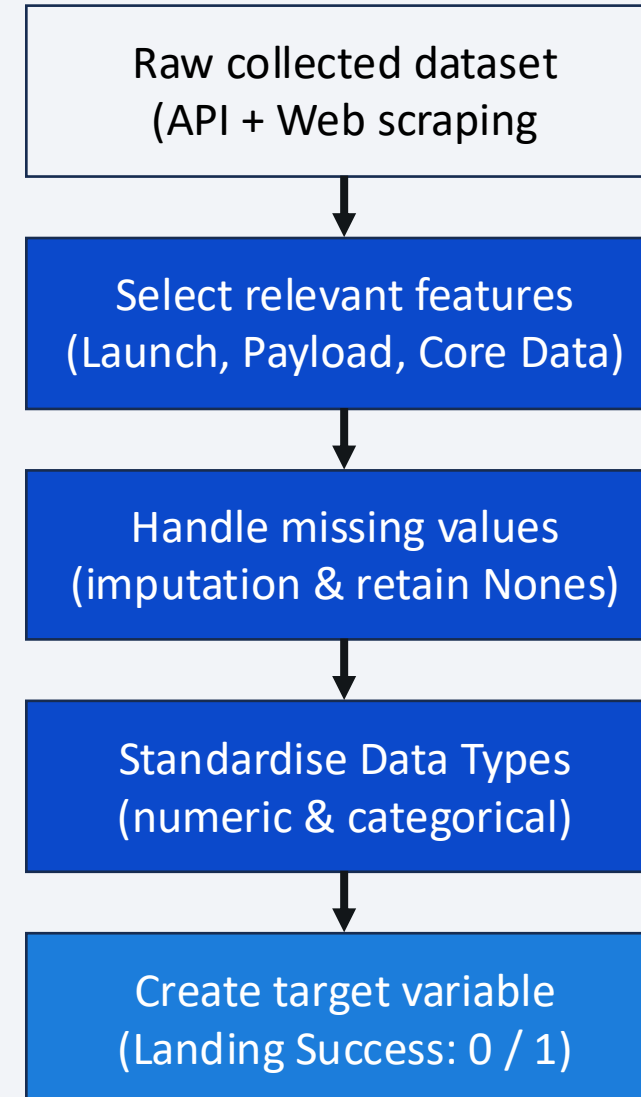
Data Collection - Scraping

- Github URL of SpaceX API notebook:
- <https://github.com/sja08/Capstone1/blob/main/10.2%20lab%20-%20web%20scraping%20beautifulsoup%20data%20collection.ipynb>



Data Wrangling

- Flow chart describes the data wrangling process
- GitHub URL data wrangling notebook
- <https://github.com/sja08/Capstone1/blob/main/10.3%20lab%20-%20data%20wrangling.ipynb>



EDA with Data Visualization

- Scatter plots (Flight Number vs. Launch Site / Payload / Orbit):
 - To identify trends in launch outcomes over time and explore relationships between payload mass, orbit type, and launch success.
- Bar charts (Success Rate by Orbit Type):
 - To compare categorical variables and assess how launch success varies across different orbit classes.
- Line chart (Yearly Average Success Rate):
 - To visualise temporal trends and evaluate improvements in Falcon 9 landing success over time.
- These visualisations helped reveal key patterns, relationships, and trends relevant to predicting first-stage landing success.

EDA with SQL

- Using bullet point format, summarize the SQL queries you performed
- Retrieved **unique** launch sites to identify all SpaceX launch locations.
- **Filtered** records using **string pattern matching** to extract launches from specific sites (e.g. CCAFS).
- Calculated total payload mass for NASA (CRS) missions using **aggregate functions**.
- Computed **average** payload mass for specific booster versions.
- Identified the first successful ground pad landing date using **MIN()** aggregation.
- Queried boosters with successful drone ship landings within a specified payload mass range.
- **Counted** successful vs. failed mission outcomes using **GROUP BY**.
- Used **subqueries** to find booster versions carrying the **maximum payload mass**.
- Extracted year- and month-specific failure records using **date substring filtering**.
- Ranked landing outcomes by frequency within a defined date range.

GitHub URL of EDA with SQL notebook: <https://github.com/sja08/Capstone1/blob/main/10.4%20lab%20-%20SQL%20with%20sqlite.ipynb>

Build an Interactive Map with Folium

- Markers:
 - Added to represent SpaceX launch sites, providing precise geographic locations and enabling interactive inspection of site details.
- Circles:
 - Used to visualise the proximity radius around launch sites, helping assess nearby infrastructure, coastlines, and populated areas relevant to launch and landing considerations.
- Polylines (Lines):
 - Plotted to represent distances between launch sites and landing locations, supporting spatial reasoning about recovery logistics and mission feasibility.
- Feature Groups and Layer Control:
 - Implemented to organise map elements into toggleable layers, improving readability and allowing users to explore individual spatial features independently.

GitHub URL of interactive map with Folium map notebook:

<https://github.com/sja08/Capstone1/blob/main/10.6%20lab%20-%20folium%20maps.ipynb>

Build a Dashboard with Plotly Dash

- Launch Site Dropdown (Interaction):
 - Added to allow users to filter results by individual launch sites or view all sites, enabling focused comparison of launch performance.
- Pie Chart (Success vs. Failure):
 - Used to visualise the proportion of successful and failed launches for a selected launch site, providing an immediate summary of launch reliability.
- Payload Mass Range Slider (Interaction):
 - Implemented to let users dynamically filter launches by payload mass, supporting exploration of how payload size impacts landing success.
- Scatter Plot (Payload Mass vs. Launch Outcome):
 - Added to examine the relationship between payload mass and launch success, with colour encoding used to distinguish successful and unsuccessful landings.
- These plots and interactive controls enable exploratory analysis by allowing users to dynamically investigate how launch site and payload mass influence first-stage landing success.

GitHub URL of Plotly Dash lab:

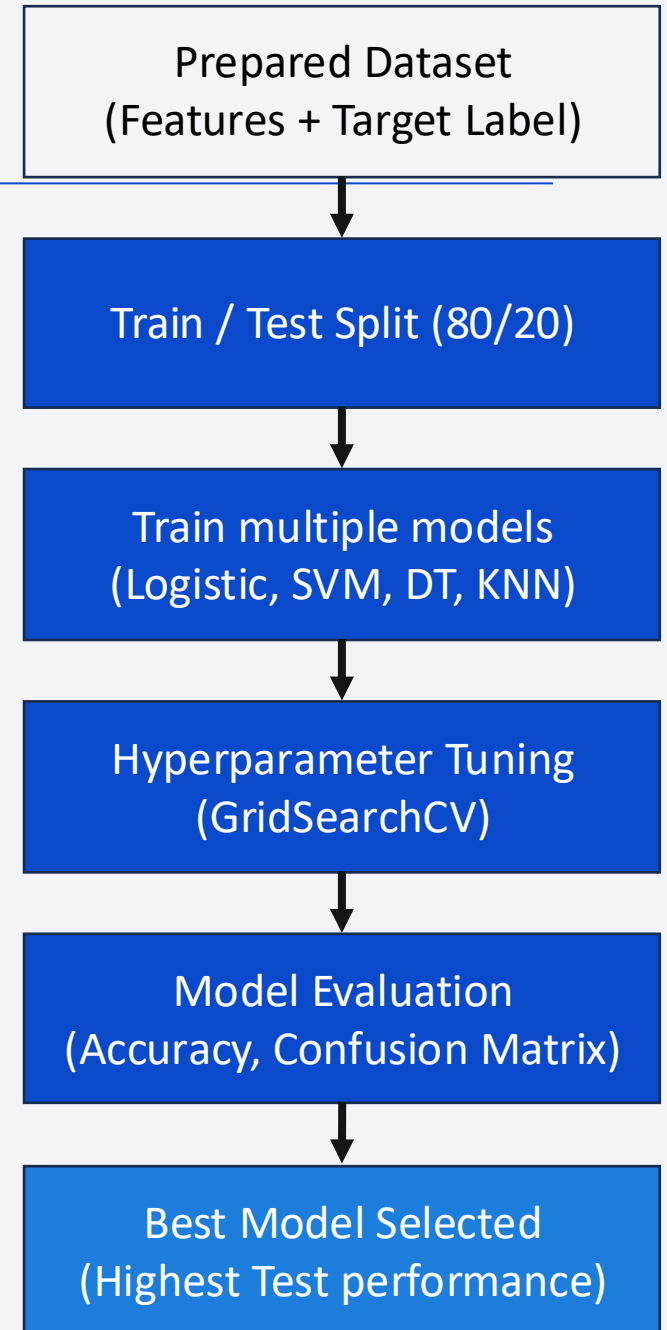
<https://github.com/sja08/Capstone1/blob/main/10.7%20lab%20-%20dashboard%20spacex-dash-app.py>

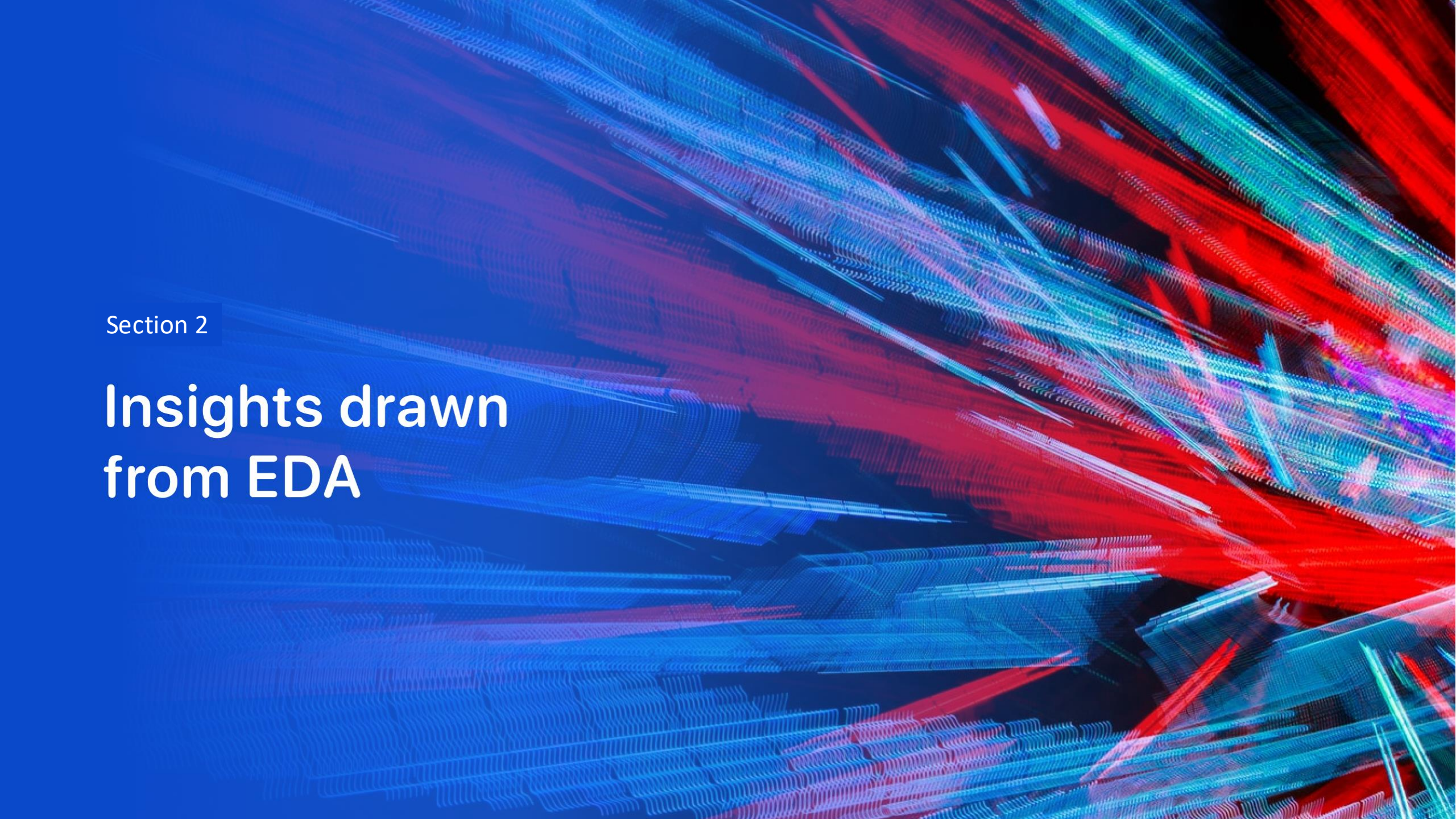
Predictive Analysis (Classification)

- **Feature selection & target definition:** Used engineered features from launch, payload, and booster data with a binary landing success label.
- **Train–test split:** Split the dataset into training and testing subsets to enable unbiased performance evaluation.
- **Model training:** Trained multiple classification algorithms including logistic regression, support vector machines, decision trees, and k-nearest neighbours.
- **Model improvement:** Applied GridSearchCV with cross-validation to tune key hyperparameters for each model.
- **Model evaluation:** Assessed performance using test accuracy and confusion matrices to compare predictive capability.
- **Model selection:** Selected the best-performing model based on highest and most consistent test performance.

GitHub URL of Predictive Analysis notebook:

<https://github.com/sja08/Capstone1/blob/main/10.8%20lab%20-%20ML%20Prediction.ipynb>



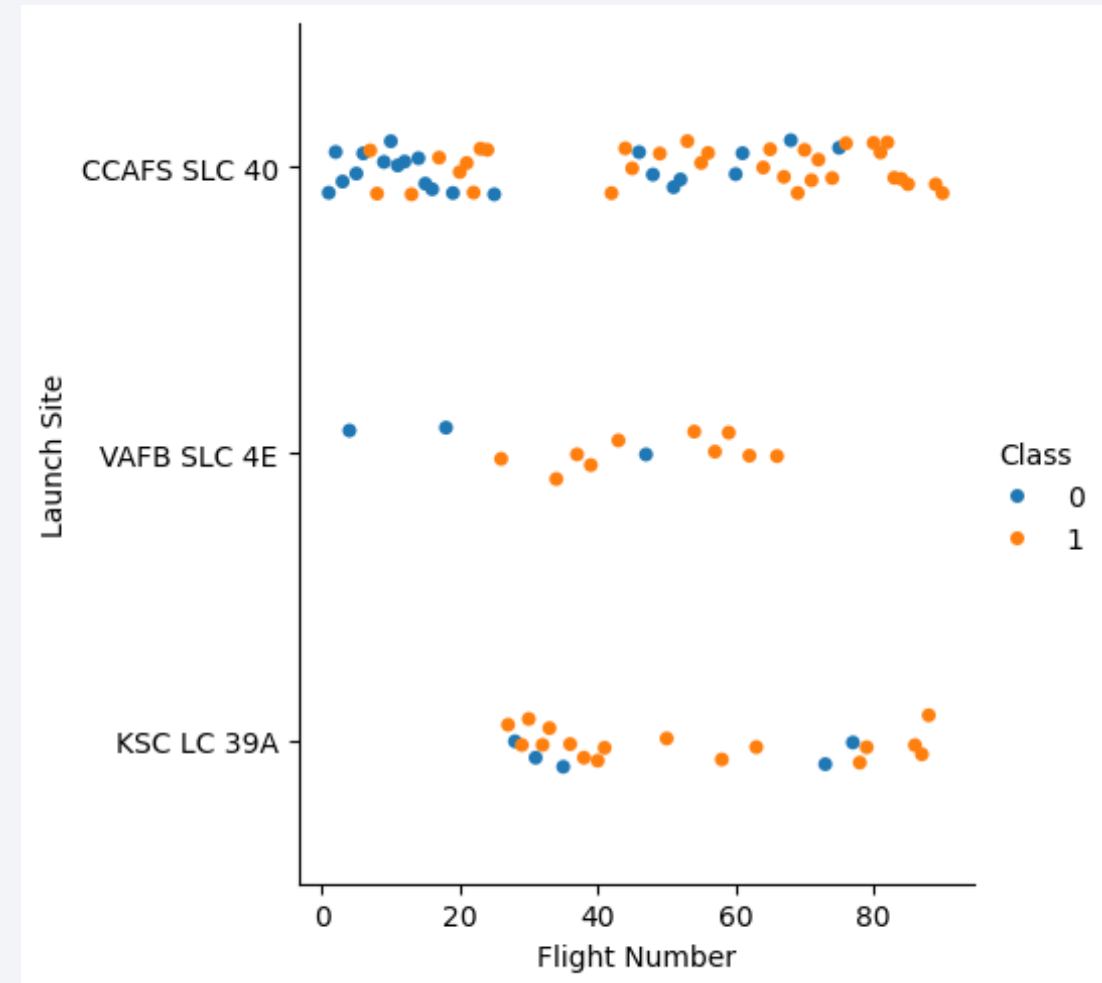
The background of the slide is an abstract composition. It features a solid blue area on the left side, which transitions into a dynamic pattern of diagonal streaks in shades of blue and red on the right. Overlaid on these streaks is a faint, white grid pattern, reminiscent of a data visualization or a technical drawing. The overall effect is one of high-tech or digital data.

Section 2

Insights drawn from EDA

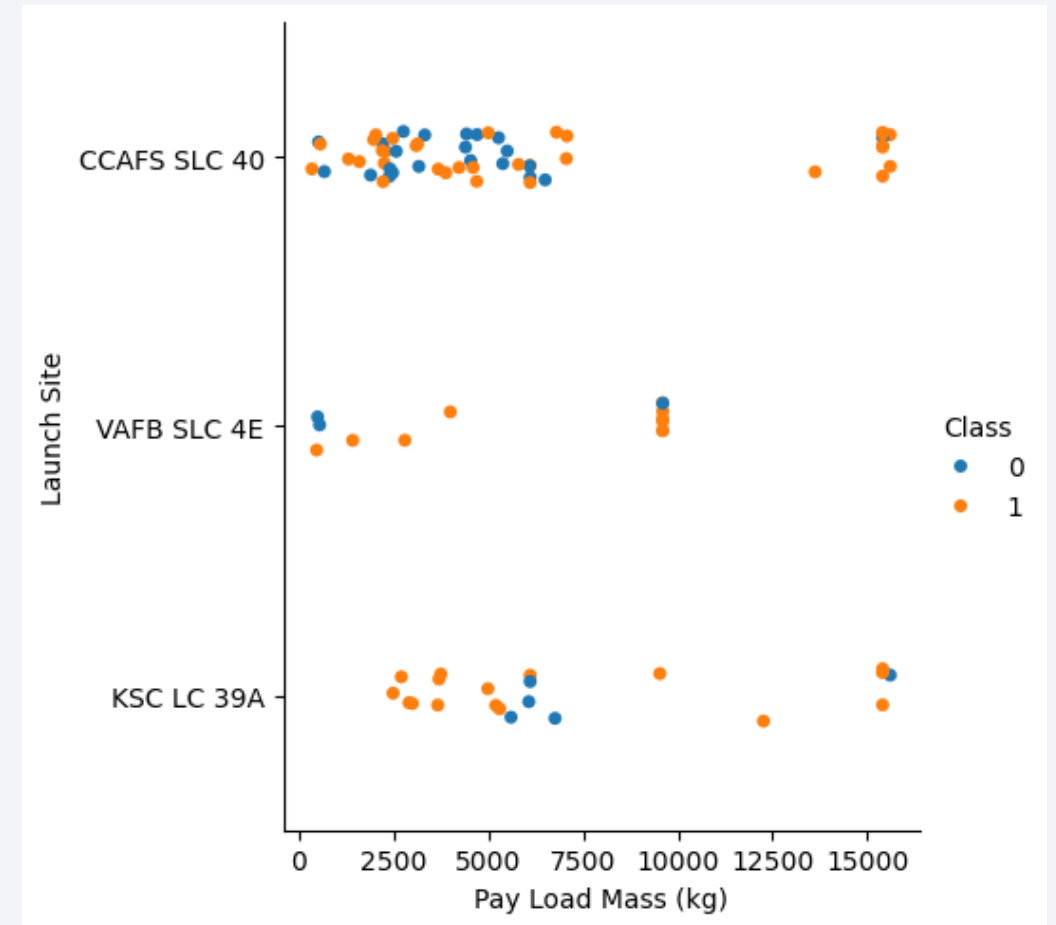
Flight Number vs. Launch Site

- This scatter plot shows launch outcomes over time (flight number) by launch site, where each point represents a Falcon 9 launch
 - The colour indicates landing outcome (Class 0 = failure, Class 1 = success).
 - It indicates that landing success increases with flight number (most prevalent for VAFB SLC 4E).
-
- Note the instructions were:
 - # Plot a scatter point chart with x axis to be Flight Number and y axis to be the launch site, and hue to be the class value



Payload vs. Launch Site

- This scatter plot shows landing outcomes by payload mass and launch site.
- Colour indicates landing outcome (Class 0 = failure, Class 1 = success).
- It shows how landing success varies with payload mass across different sites, helping assess whether heavier payloads are associated with lower success rates and whether this relationship differs by launch location.
- For the VAFB-SLC launchsite there are no rockets launched for a heavy payload mass (greater than 10,000 kg).



Success Rate vs. Orbit Type

Highest success rates are observed for:

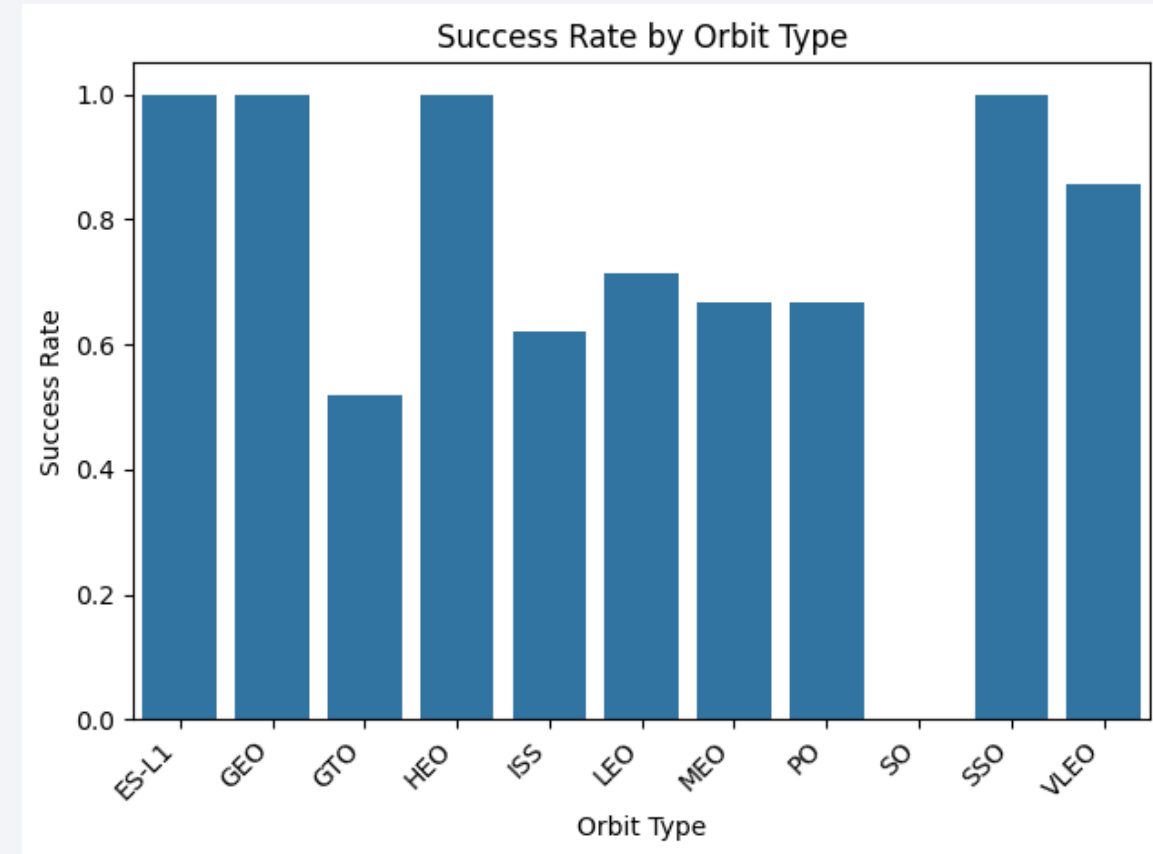
- ES-L1, GEO, HEO, and SSO → ~100% success rate
 - consistently successful first-stage landings, likely due to more mature mission profiles and operational experience.
- VLEO → High success rate (~85–90%)
 - Indicates strong landing performance despite varying payload and mission characteristics.

Moderate success rates

- LEO, ISS, MEO, and PO → ~60–70% success rate
 - These orbits include a wider range of missions (especially ISS and LEO), including early flights where landing technology was still being refined.

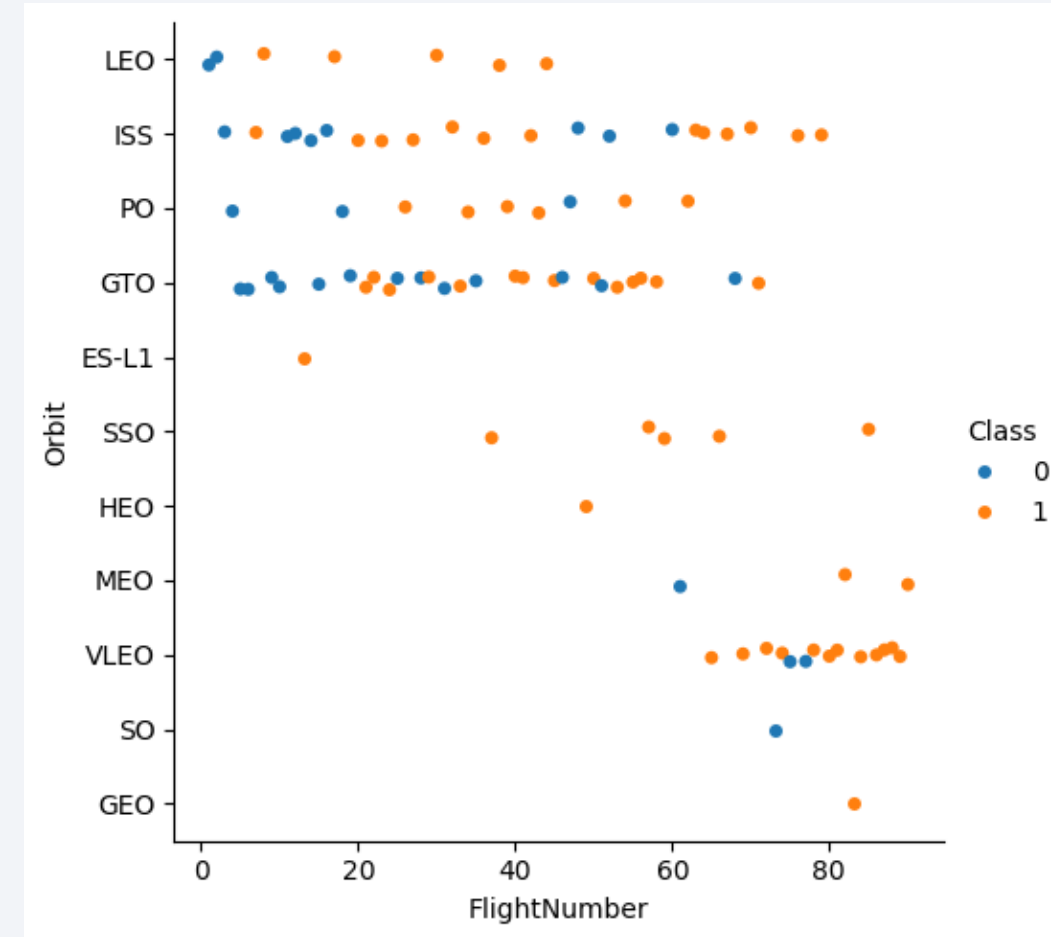
Lower success rate

- GTO → ~50% success rate
 - GTO missions typically involve heavier payloads and higher energy requirements, making booster recovery more challenging.



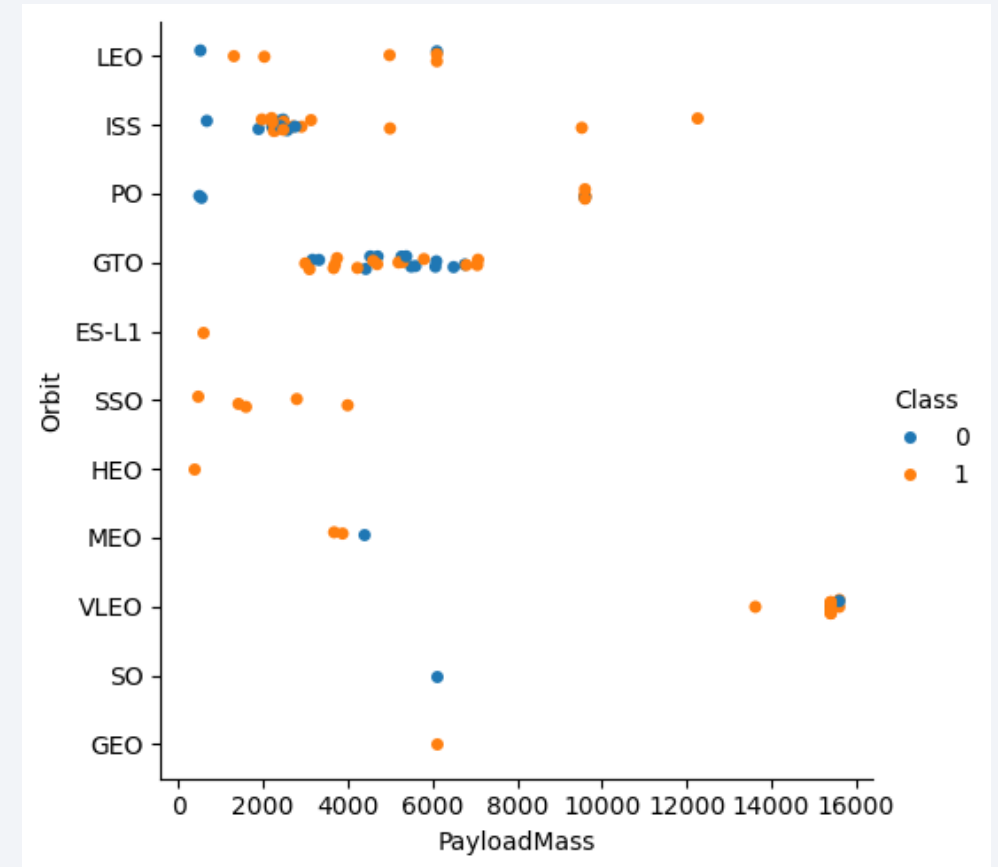
Flight Number vs. Orbit Type

- Shows landing outcomes by orbit type over time
- Colour indicates landing outcome (Class 0 = failure, Class 1 = success).
- Early missions across several orbits (e.g. LEO, ISS, GTO) include more failures
- Later flights result in more successful landings (especially VLEO, SSO, GEO, and ES-L1)
- This highlights both technological improvement over time and possibly how mission orbit influences landing success.



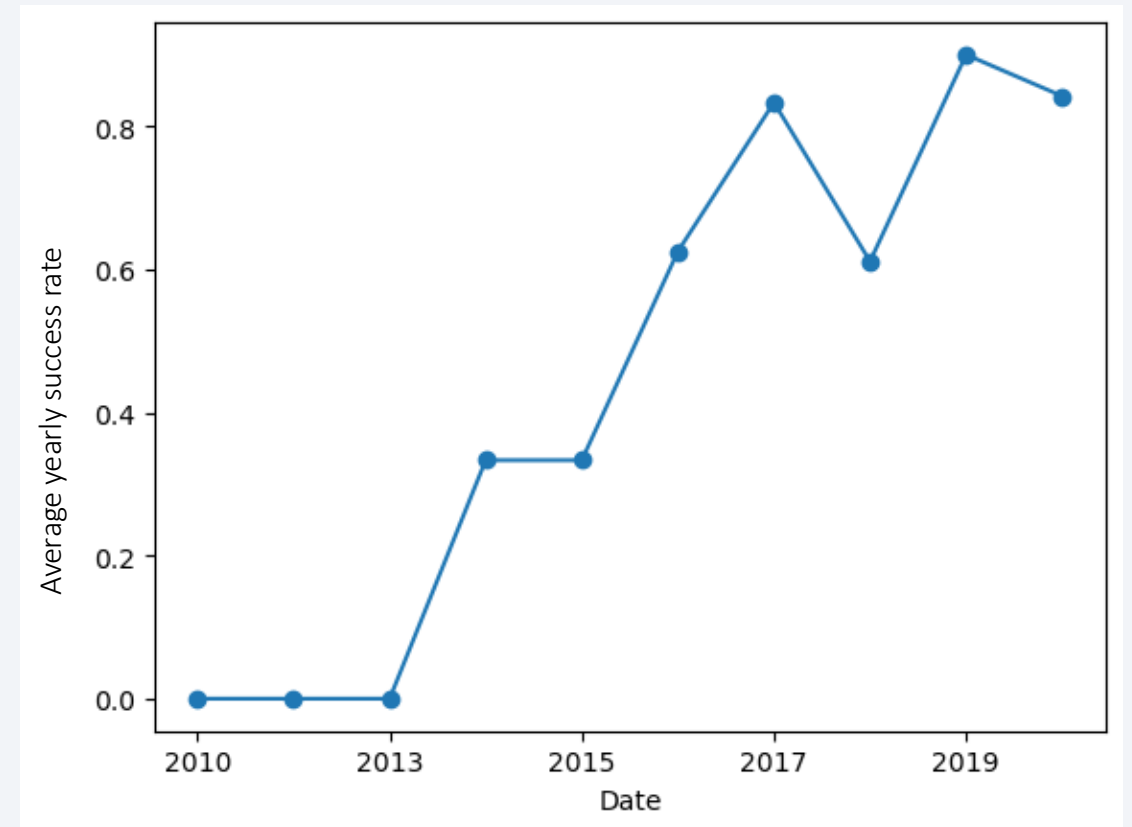
Payload vs. Orbit Type

- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.
- However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.



Launch Success Yearly Trend

- Success rate since 2013 kept increasing till 2020, with a decrease in 2018 and 2020 from the previous year.



All Launch Site Names

- Query uses DISTINCT to return only the unique values from the column
- There are 4 launch sites, as listed in the image to the right.

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db  
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- A filter using the WHERE query was applied to only show results that begin with CCA
- As only 5 records were requested, the return was limited to 5 using the LIMIT query.

```
%sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db  
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Missio
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	

Total Payload Mass for boosters from NASA

- Calculate the total payload carried by boosters from NASA
 - Use of WHERE query to filter for boosters from NASA
 - Use of the SUM() function to sum up the masses to give a total.
 - The total payload mass carried by NASA boosters is 45,596 kg

```
: %sql SELECT SUM("PAYLOAD_MASS__KG_") as Total_Payload_Mass FROM SPACEXTABLE WHERE Customer = 'N
* sqlite:///my_data1.db
Done.
: Total_Payload_Mass
    45596
```


Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1
 - Use of WHERE to filter the specific booster version
 - Use of the AVG() function to calculate the average
 - The average payload mass for the F9 v1.1 booster is 2928.4 kg

```
%sql SELECT AVG("PAYLOAD_MASS__KG_") as Avg_payload_mass FROM SPACEXTABLE WHERE "Booster_Versio
```

```
* sqlite:///my_data1.db
```

```
Done.
```

<u>Avg_payload_mass</u>

2928.4

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad
 - Used WHERE to filter for all successful landings on ground pad
 - Used the MIN() function to select the earliest date, representative of the first successful ground landing date.
 - The first successful ground landing was on 22/12/2015.

```
%sql SELECT MIN("Date") as First_Successful_Ground_Landing FROM SPACEXTABLE WHERE "Landing_Outc
```

```
* sqlite:///my_data1.db  
Done.
```

```
First_Successful_Ground_Landing
```

```
2015-12-22
```

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000
 - Used WHERE coupled with AND conditionals to filter for the requirements.
 - 4 booster versions met the criteria.

```
%sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = "Success (drone ship)" AND "PAYLOAD_MASS_KG_" > 4000 AND "PAYLOAD_MASS_KG_" <
```

Python ▾

```
* sqlite:///my\_data1.db
```

Done.

Booster_Version

F9 FT B1022

F9 FT B1026

F9 FT B1021.2

F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes
- Present your query result with a short explanation here
- We can see that the majority of the mission outcomes were a success

```
%sql SELECT "Mission_Outcome", COUNT(*) as Total_Count FROM SPACEXTABLE GROUP BY "Mission_Outcome";
```

* [sqlite:///my_data1.db](#)

Done.

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
 - The inner query (SELECT MAX(...)) calculates the largest payload mass recorded in the table.
 - The outer query then returns the Booster_Version for all launches where the payload mass equals that maximum value.
- Multiple Falcon 9 Block 5 boosters were used on missions carrying the heaviest payload mass.

```
%sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG" = (SELECT MAX("PAYLOAD_MASS_KG") FROM SPACEXTABLE);
```

```
* sqlite:///my\_data1.db  
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015
 - In 2015, drone ship landing failures occurred twice, both launched from CCAFS LC-40, involving Falcon 9 v1.1 boosters B1012 and B1015.

```
%sql SELECT substr(Date, 6, 2) as Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date, 1, 4) = '2015' AND Landing_Outcome = 'Failure (drone ship)'
```

```
* sqlite:///my\_data1.db
```

Done.

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
 - Filters** launches within the specified date range and **groups** data by Landing_Outcome
 - Uses **COUNT(*)** to calculate frequency and **sorts** results in descending order so the most common outcome appears first
- Early missions had no landing attempt, while drone ship landings were the most common attempted recovery method, with similar numbers of successes and failures in this time period.

```
%sql SELECT Landing_Outcome, COUNT(*) as Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Landing_Outcome ORDER BY Count
```

* [sqlite:///my_data1.db](#)
Done.

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

Global Map of SpaceX Launch Sites

- This map shows the location of the launch sites on a global map.
- Launch sites are concentrated in the United States, mainly:
 - Florida (Cape Canaveral / Kennedy Space Center) on the east coast
 - California (Vandenberg) on the west coast



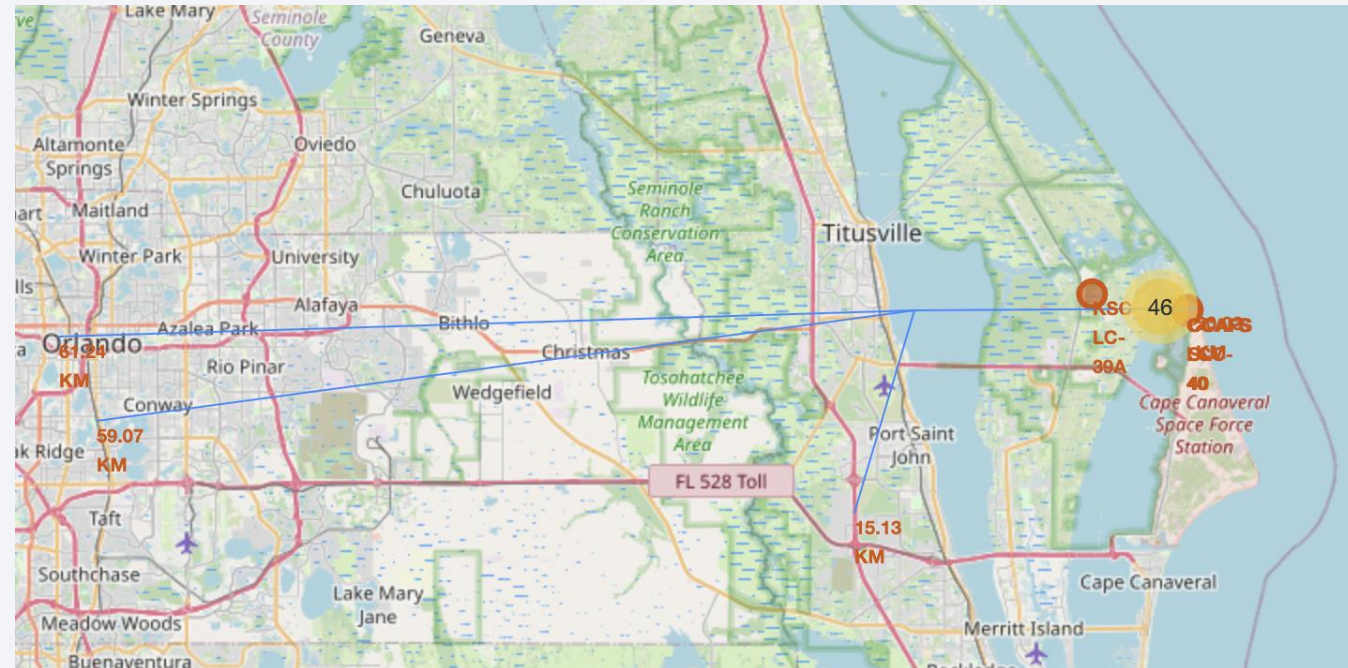
Launch Outcomes by Location

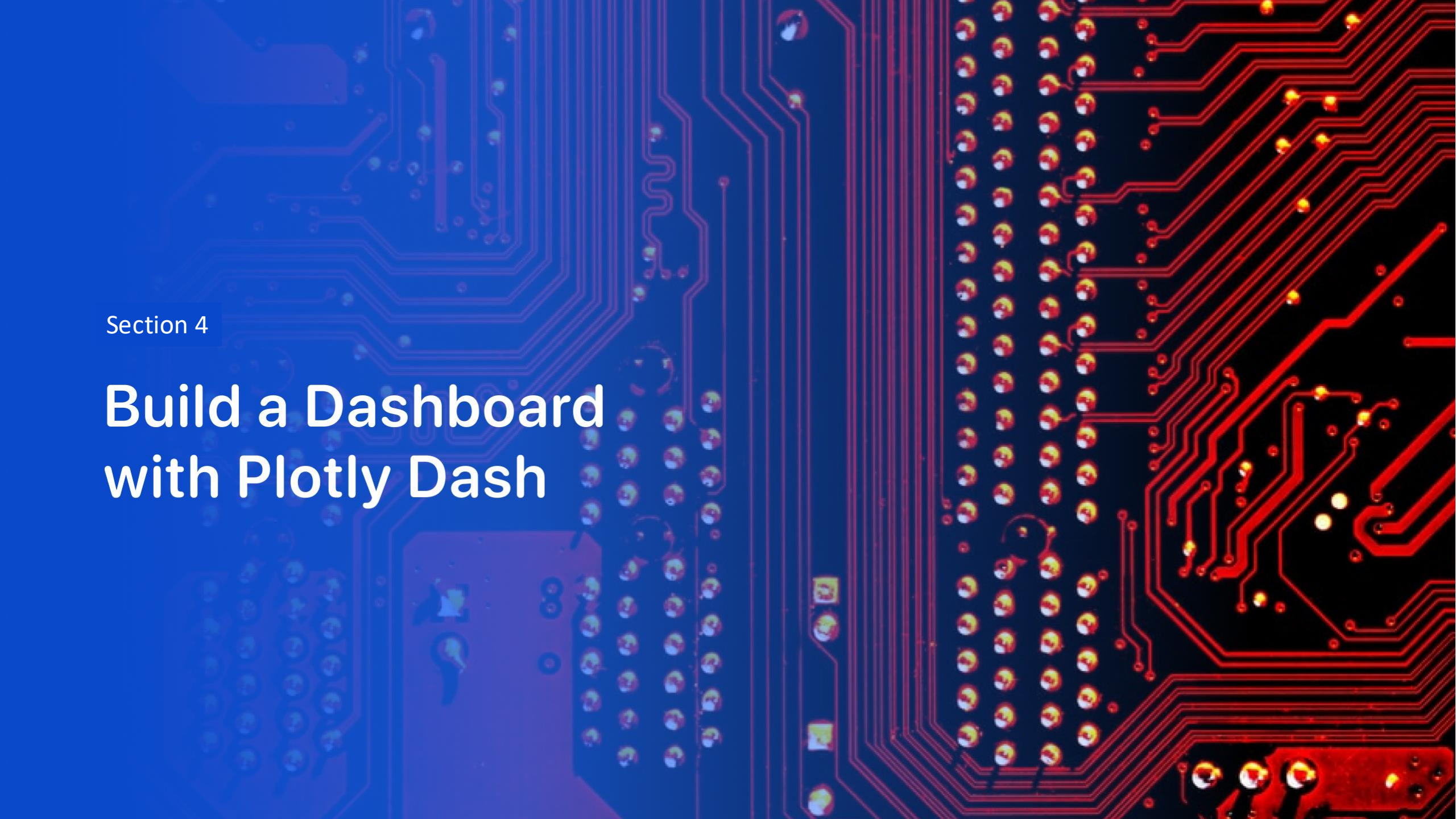
- The map shows the color-labeled launch outcome(s) for a given launch site when you zoom in and click on the site. As you zoom in, the number of launches will be displayed.
- Colour-coded markers represent landing outcomes:
 - **Green** = Successful landing (Class 1)
 - **Red** = Unsuccessful landing (Class 0)
- Market clustering helps manage overlapping points and improves readability.
- Popups/tooltips allow interactive exploration of outcome details for specific launches.



Proximity Analysis for a Selected Launch Site

- Proximity of a launch site to a chosen geographical feature is shown on this map.
- The chosen geographical features are a City, Railway, Highway and Coast
- This demonstrates the power of these maps in providing useful information, such as calculating the distance between geographical features and showing the outcome and lines on the map.



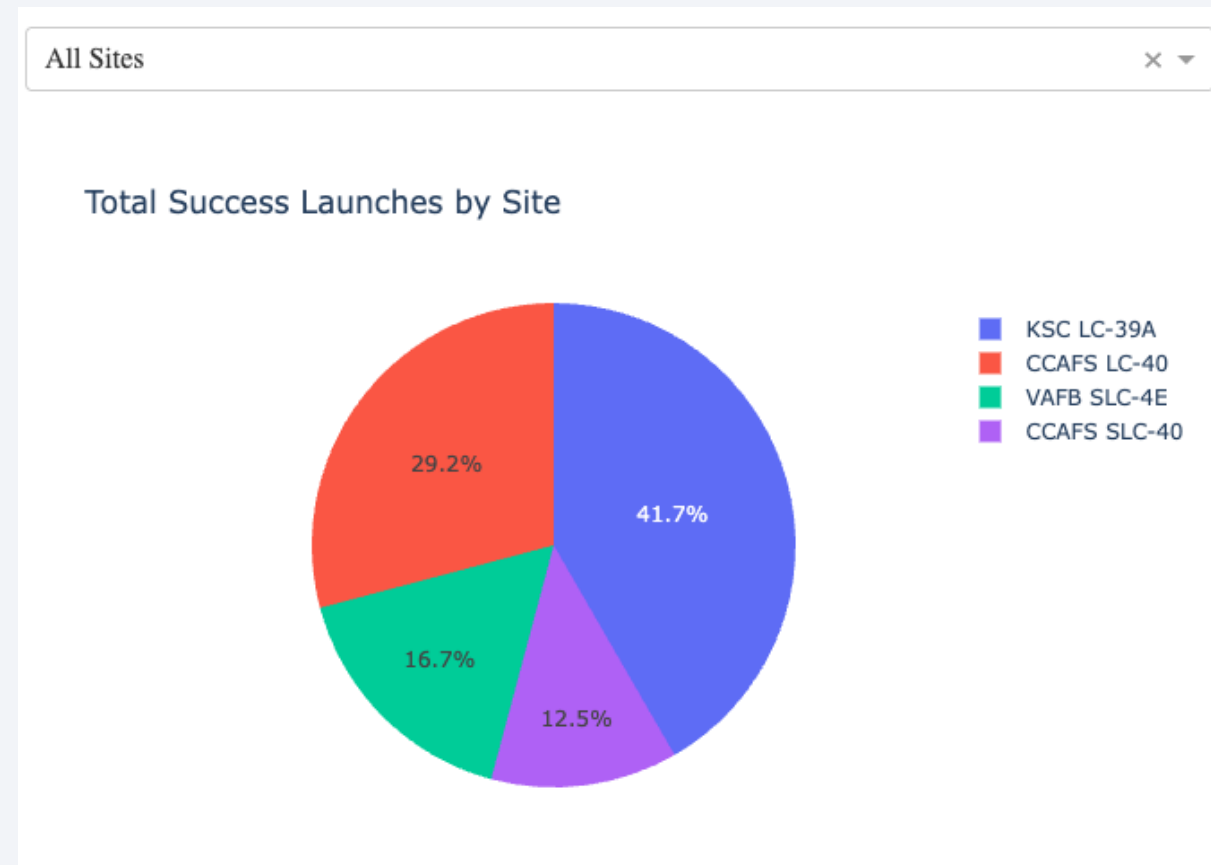


Section 4

Build a Dashboard with Plotly Dash

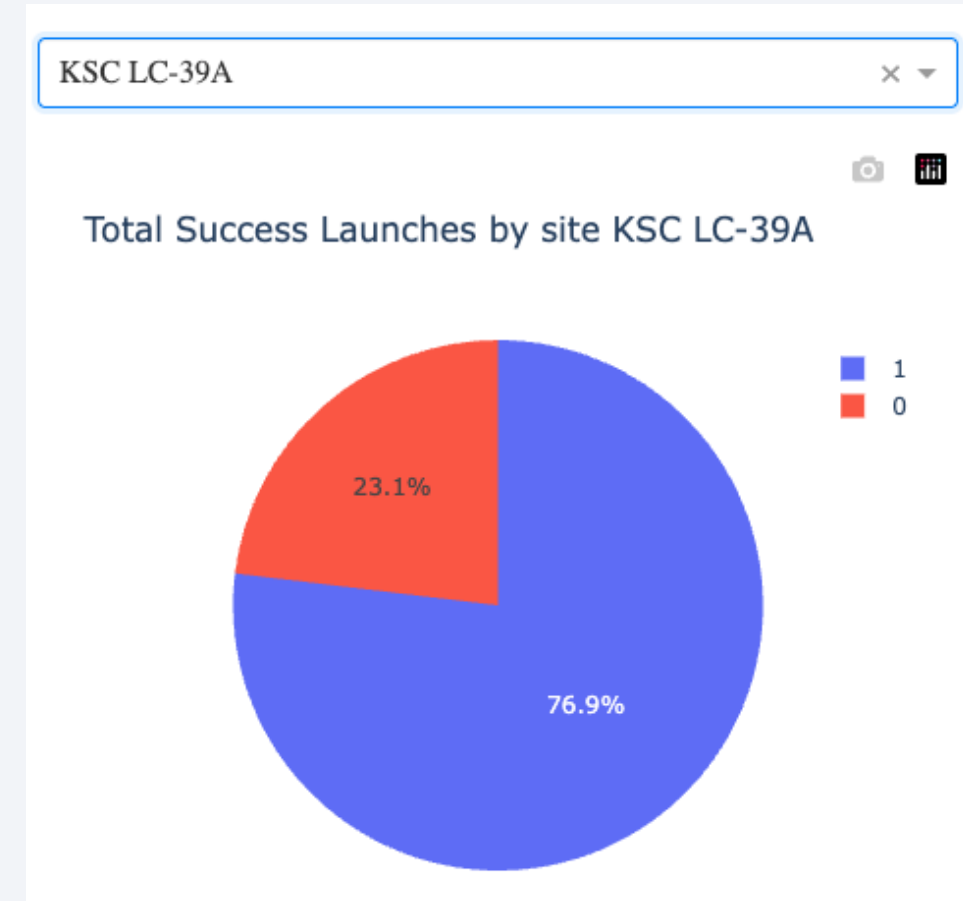
Distribution of Successful Launches by Site

- This pie chart shows the distribution of successful launches by launch site
- From the chart:
 - KSC LC-39A accounts for the largest share of successful launches (~41.7%)
 - CCAFS LC-40 is next with ~29.2%
 - VAFB SLC-4E contributes ~16.7%
 - CCAFS SLC-40 contributes the smallest share at ~12.5%
- This indicates that most successful launches in the dataset occurred from KSC LC-39A and CCAFS LC-40, suggesting these sites are SpaceX's primary operational launch locations (in this dataset).



Launch Success Ratio for Highest-Performing Site

- Use of the dropdown feature on the dashboard allows the launch site with the highest success rate to be easily evaluated.
- The site with the highest success rate is:
 - KSC LC-39A at 76.9% successful launches.



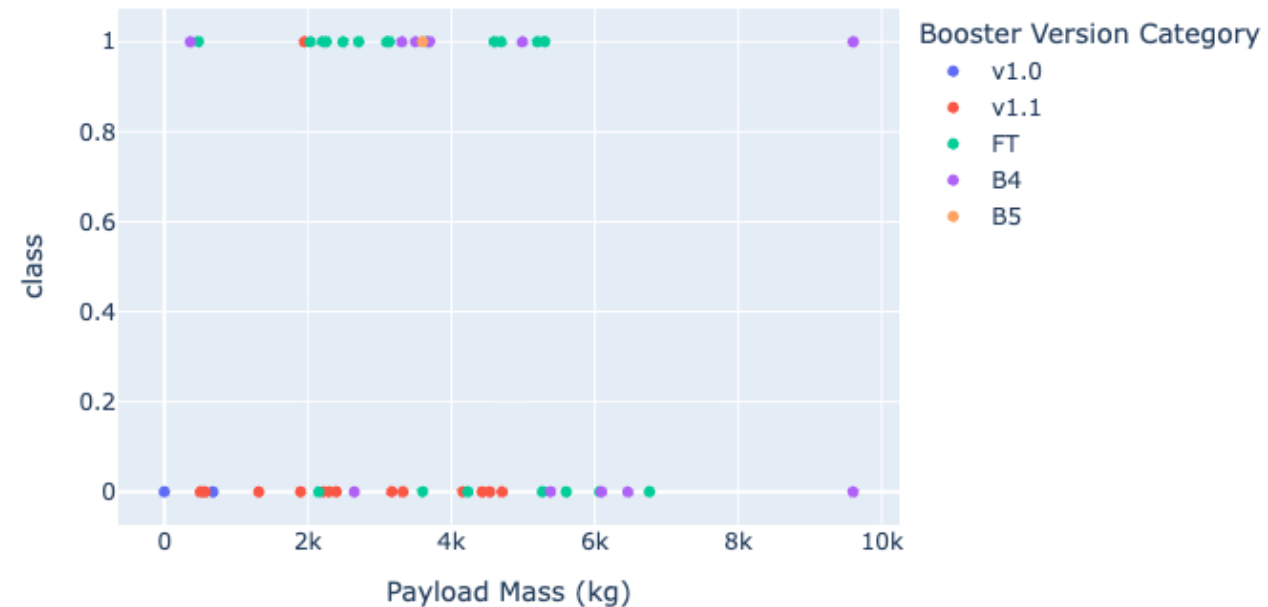
Payload Mass vs Launch Success (All Sites)

- Launch outcomes are mixed across most payload masses, meaning payload mass alone does not perfectly determine success.
- Most failures (class = 0) are concentrated in the lower-to-mid payload range (~0–5000 kg), while many successes occur across a broader range.
- Newer booster categories (FT/B4/B5) appear more frequently among successful outcomes, suggesting improved performance/reusability compared with early versions (v1.0/v1.1).

Payload range (Kg):



Correlation between Payload and Success for all Sites



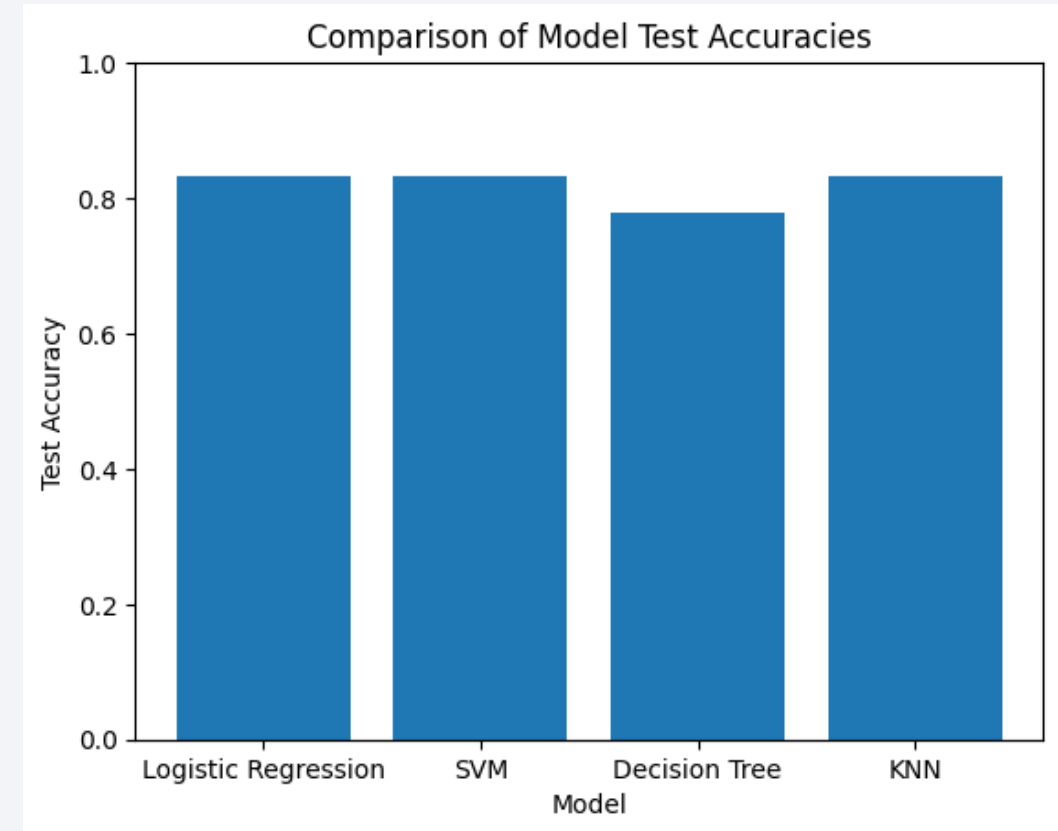


Section 5

Predictive Analysis (Classification)

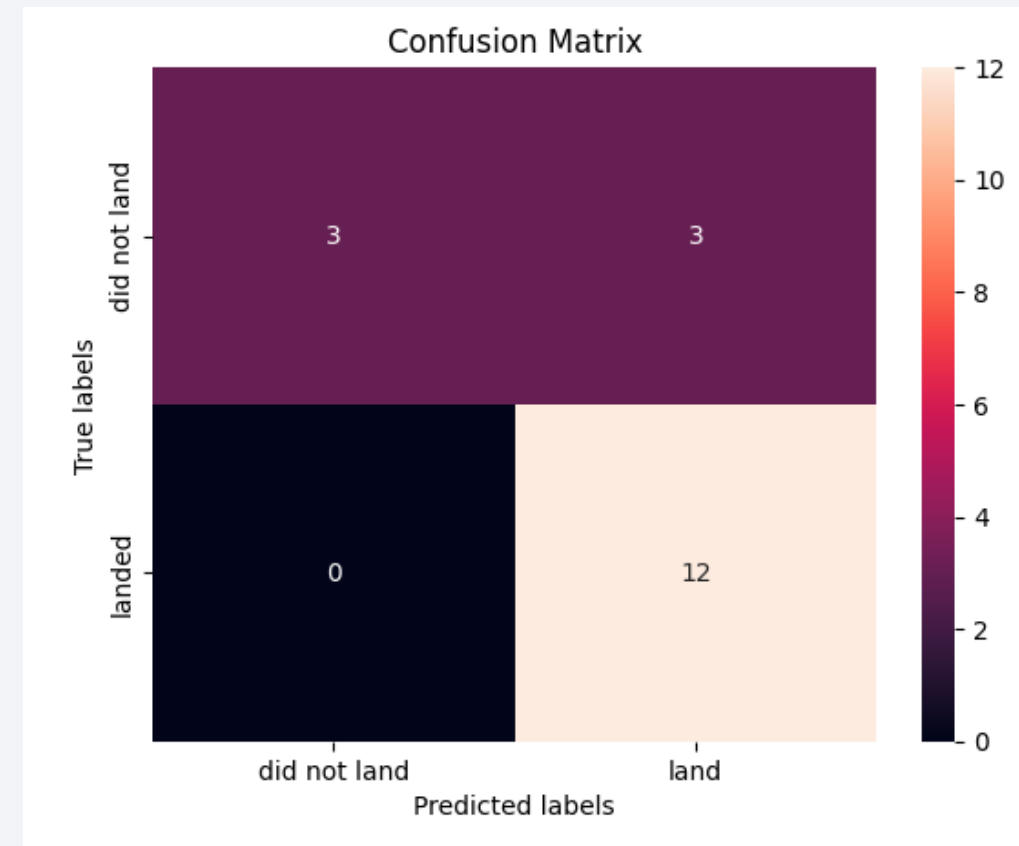
Classification Accuracy

- This bar chart shows the accuracy for the classification models.
- Logistic Regression, SVM and KNN have the highest overall accuracies.



Confusion Matrix

- This confusion matrix was the same for Logistic Regression, SVM and KNN.
- Explanation of confusion matrix:
 - **12 true positives:** launches that landed were correctly predicted as “landed”.
 - **3 true negatives:** launches that did not land were correctly predicted as “did not land”.
 - **3 false positives:** launches that did not land were incorrectly predicted as “landed”.
 - **0 false negatives:** no launches that landed were incorrectly predicted as “did not land”.
 - Perfect **recall** for the “landed” class (the model did not miss any successful landings).
 - Some over-prediction of landings, leading to a small number of false positives.
 - Overall accuracy = $15 / 18 \approx 83.3\%$.



Introduction

- **Context:** SpaceX has dramatically reduced launch costs by reusing the Falcon 9 first stage, giving it a strong competitive advantage.
- **Challenge:** For a new launch company (Space Y), launch pricing depends heavily on whether the first stage can be recovered and reused.
- **Problems to be answered:**
 1. What factors (e.g. payload mass, orbit type, launch site) influence whether the Falcon 9 first stage successfully lands?
 2. Can we predict first-stage landing success using publicly available launch data?
 3. Which machine learning model provides the most accurate and reliable predictions?
 4. How can these predictions be used to estimate launch cost and support pricing decisions?
- **Goal:** Develop a data-driven model and dashboards to support cost estimation and decision-making for Space Y.

Conclusions

- **Key factors influencing success:** Landing success is strongly influenced by payload mass, orbit type, launch site, and flight number/booster version (reflecting improving performance over time).
- **Predictability using public data:** Using publicly available SpaceX launch data, landing success can be predicted with good accuracy using classification models.
- **Best-performing model:** After training and tuning multiple classifiers, the Logistic Regression, SVM and KNN models were found to give the same overall accuracy and confusion matrices.
- **Business value (cost/pricing):** These predictions estimate the probability of first-stage reuse, improving launch cost forecasts and supporting more informed pricing and competitive bidding decisions.

Appendix

- Please refer to the Github for all documents, raw datafiles and reference material used in this project.
- **Github:** <https://github.com/sja08/Capstone1/tree/main>

Thank you!

